

SHUVAM PAL

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Education

Dayananda Sagar College of Engineering

Bengaluru, India

B.E. Artificial Intelligence and Machine Learning

Experience

Bentham AI

Dec 2025 – Feb 2026

Backend Engineer Intern: Node.js, JavaScript, Go, Docker, GCP, GitHub Actions

Remote

- Built backend automation for compliance-sensitive MCA filings with browser execution, public-site API lookups, retries, **session recovery**, and operator checkpoints, reducing manual filing time by **85%**.
- Built draft-generation flow with inspectable outputs before deterministic filing workflow execution.
- Shipped Dockerized services on **Google Cloud Run** with **GitHub Actions** release workflows, improving deployment velocity by **70%**.

Google Summer of Code

May 2025 – Sept 2025

Mentee at Invesalious: Python, VTK, wxPython, NumPy, PyTorch, ONNX

Remote

- Integrated a **FastSurfer CNN segmentation** feature for **95 anatomical brain regions**, wiring **PyTorch/TorchScript** and **ONNX/Tinygrad** inference, plane-specific axial/coronal/sagittal models, and device selection into Invesalious.
- Built ML preprocessing and postprocessing for **NIfTI/MGZ MRI data**, including voxel conformation, thick-slice datasets, multi-view aggregation, sagittal label remapping, LUT mapping, generated **binary masks**, and quick QC checks.
- Reduced runtime by **87%** with async/parallel execution while debugging orientation, label correctness, GPU/CPU paths, progress reporting, and 3D inspection flows.

Open Source

[Contributions](#)

- Contributed patches across **Invesalious**, **Kuadrant MCP Gateway**, **Kyverno**, **Tiled**, **CRIU**, and **VideoLAN**, spanning **data/artifact correctness**, backend failover behavior, security validation, and compatibility fixes.

Projects

Hyoka | Python, FastAPI, PostgreSQL, SQLAlchemy, Next.js, TypeScript, Docker

- Built an **audit-ready AI reliability control plane** with trace/event ingestion, eval/replay workers, release gates, promotions, signed manifests, API keys, **audit logs**, and a **Next.js dashboard**.
- Designed **PostgreSQL/SQLite metadata models** for traces, suites, candidates, validation runs, gate decisions, promotions, artifacts, manifests, and project-scoped API keys using **SQLAlchemy/Alembic**.
- Implemented **database-backed worker leases**, content-addressed artifacts, OTLP/OpenInference ingestion, an OpenAI-compatible proxy, and Docker Compose services for API, worker, proxy, dashboard, and Postgres.

Swish Support System | Go, Python, FastAPI, React, PostgreSQL, Redis, Langfuse

[GitHub](#)

- Built an **operational decision workflow** that pulls order, kitchen, fleet, trust, and evidence context into LLM assessment while deterministic policy code controls refunds, replacements, coupons, and escalations.
- Modeled support cases as **policy-bound workflow state**, including evidence strength, issue type, active item, trust score, fault owner, desired resolution, and escalation path instead of relying on prompt-only decisions.

Postificus | Go, Echo, React, Vite, PostgreSQL, Redis, RabbitMQ, Docker, Prometheus

[GitHub](#)

- Built a **full-stack workflow app** with REST APIs, auth/settings endpoints, PostgreSQL persistence, Redis cache/autosave, S3-compatible storage, RabbitMQ jobs, and real-time publication status.
- Implemented **direct API plus browser-fallback execution** for fragmented third-party platforms, matching Rimba-style problems where external operational data and systems are inconsistent.
- Added **retries**, **DLQs**, **circuit breakers**, **health checks**, and **Prometheus metrics** so long-running API/browser workflows stayed observable and recoverable under unreliable integrations.

Technical Skills

Languages: TypeScript, JavaScript, Python, Go

Frameworks: React, Next.js, FastAPI, Node.js, Echo, Gin, SQLAlchemy

Tools: PostgreSQL, SQLite, Redis, RabbitMQ, Docker, GCP Cloud Run, GitHub Actions, OpenTelemetry, Prometheus, Langfuse, Codex